Reactor Performance Report

CO2-to-methanol - a figure-size catalogue

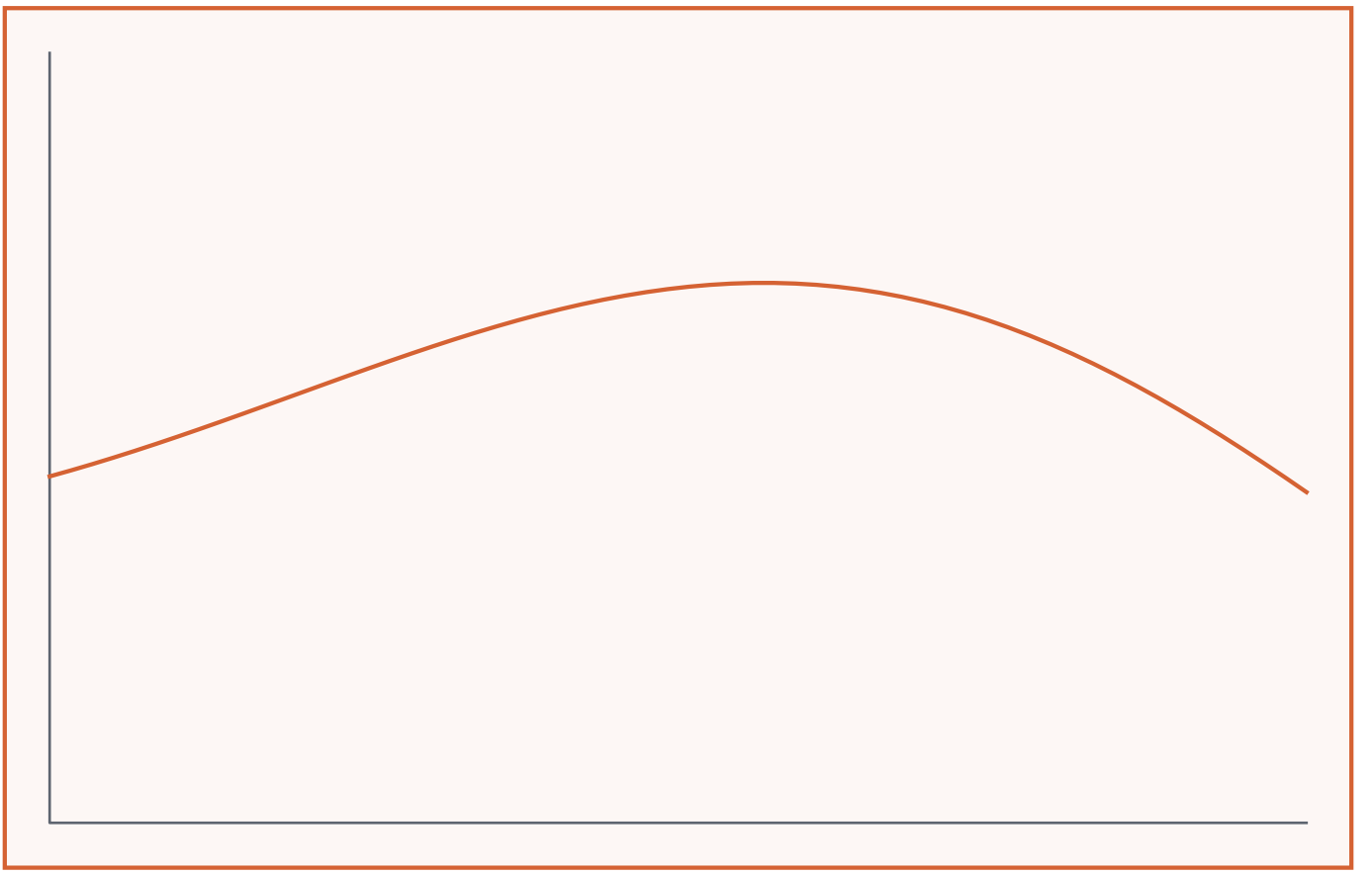


Figure 1 - double (18 x 11.5 cm)

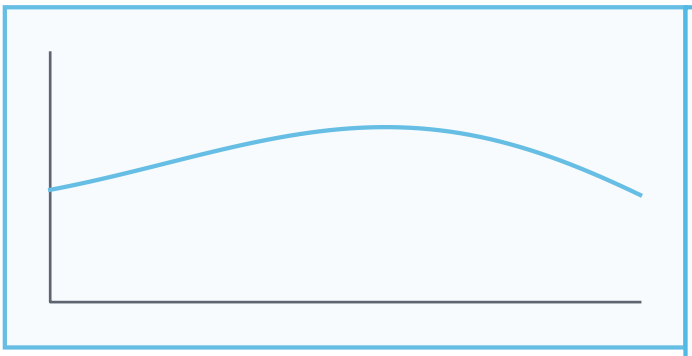


Figure 2 - single_short (9 x 4.5 cm)

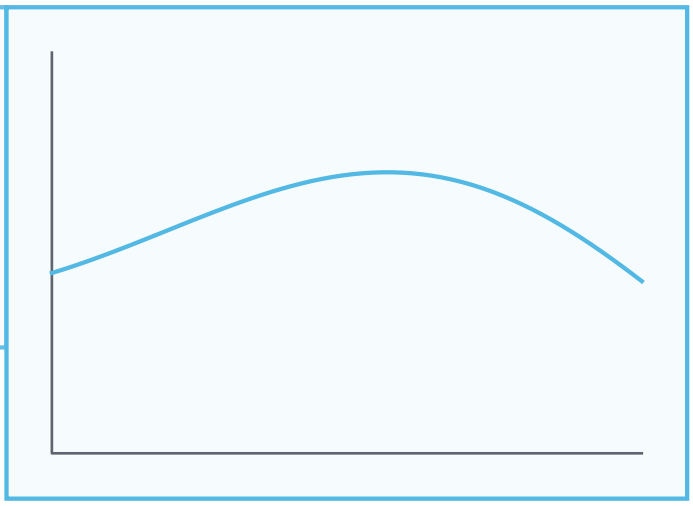


Figure 3 - single (9 x 6.5 cm)

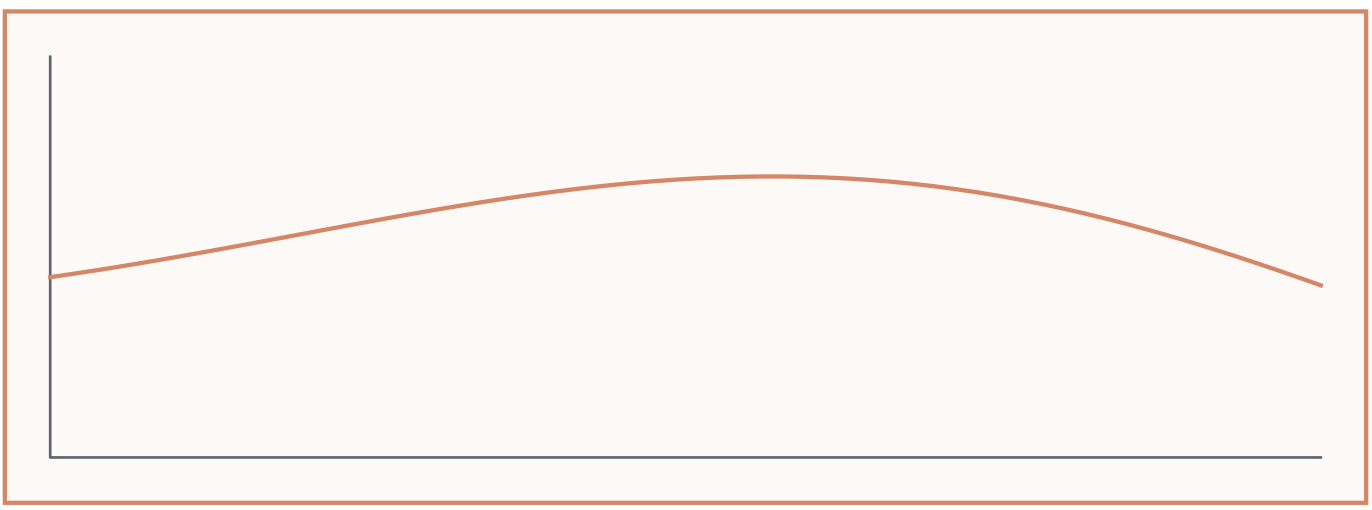


Figure 4 - double_single (18 x 6.5 cm)

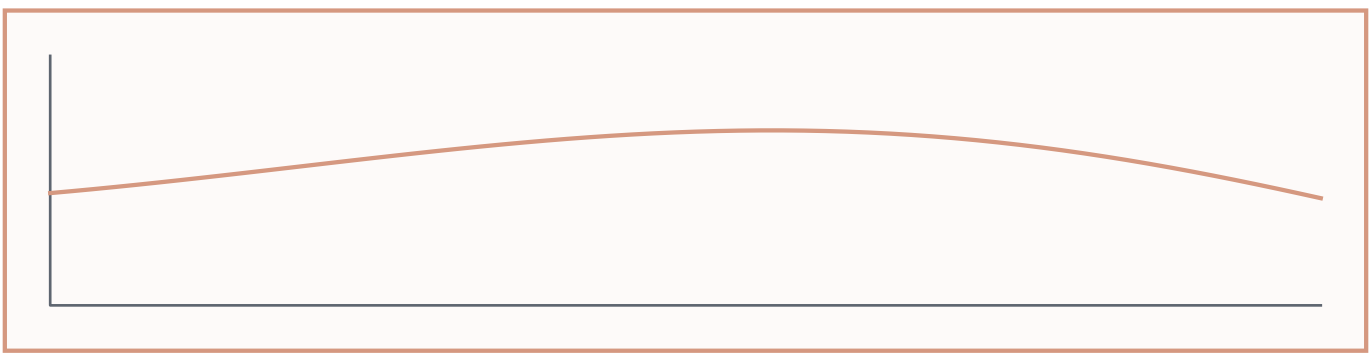


Figure 5 - double_short (18 x 4.5 cm)

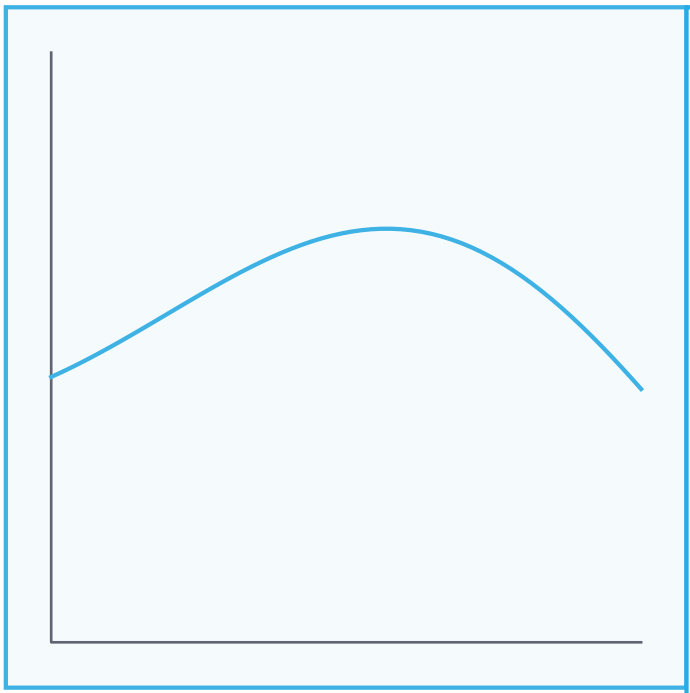


Figure 6 - single_square (9 x 9 cm)

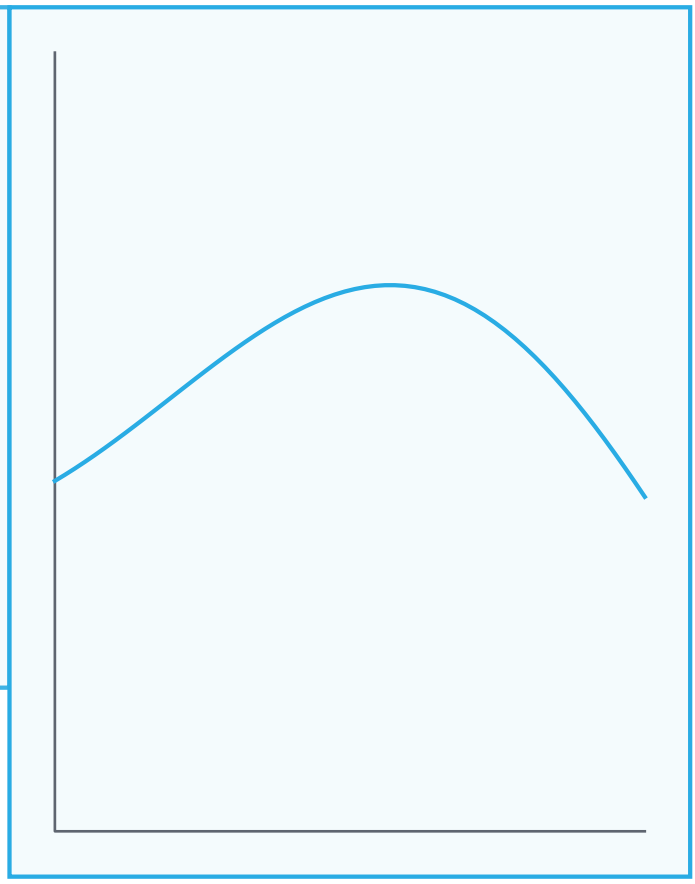


Figure 7 - single_double (9 x 11.5 cm)

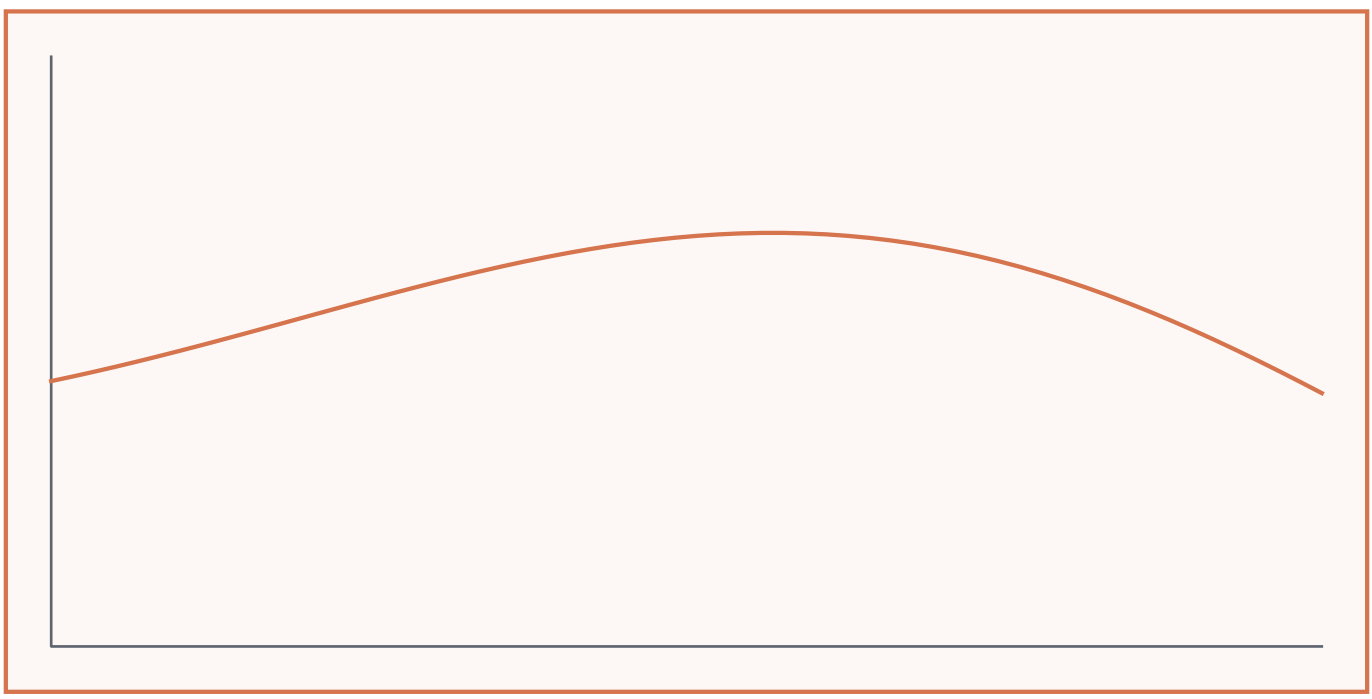


Figure 8 - double_single_square (18 x 9 cm)

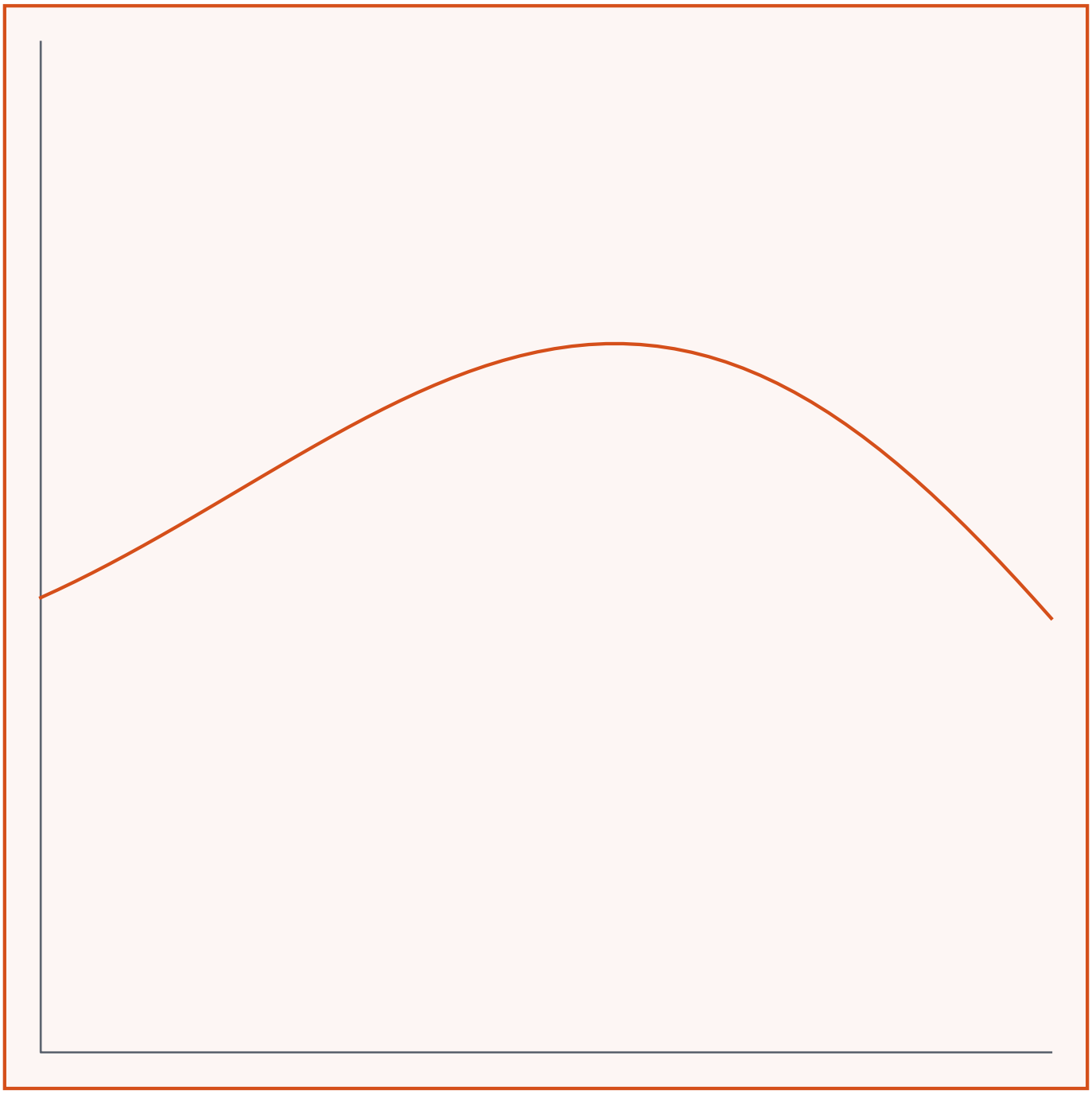


Figure 9 - double_square (18 x 18 cm)

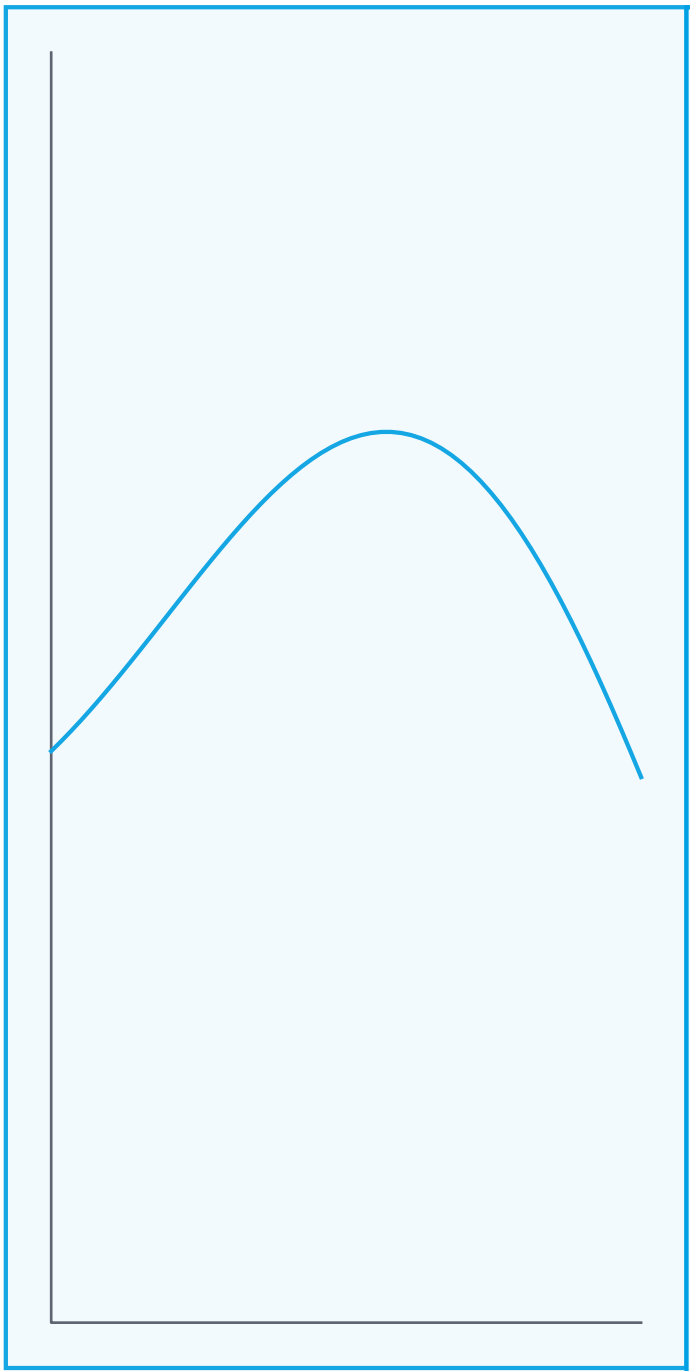


Figure 10 - single_double_square (9 x 18 cm)

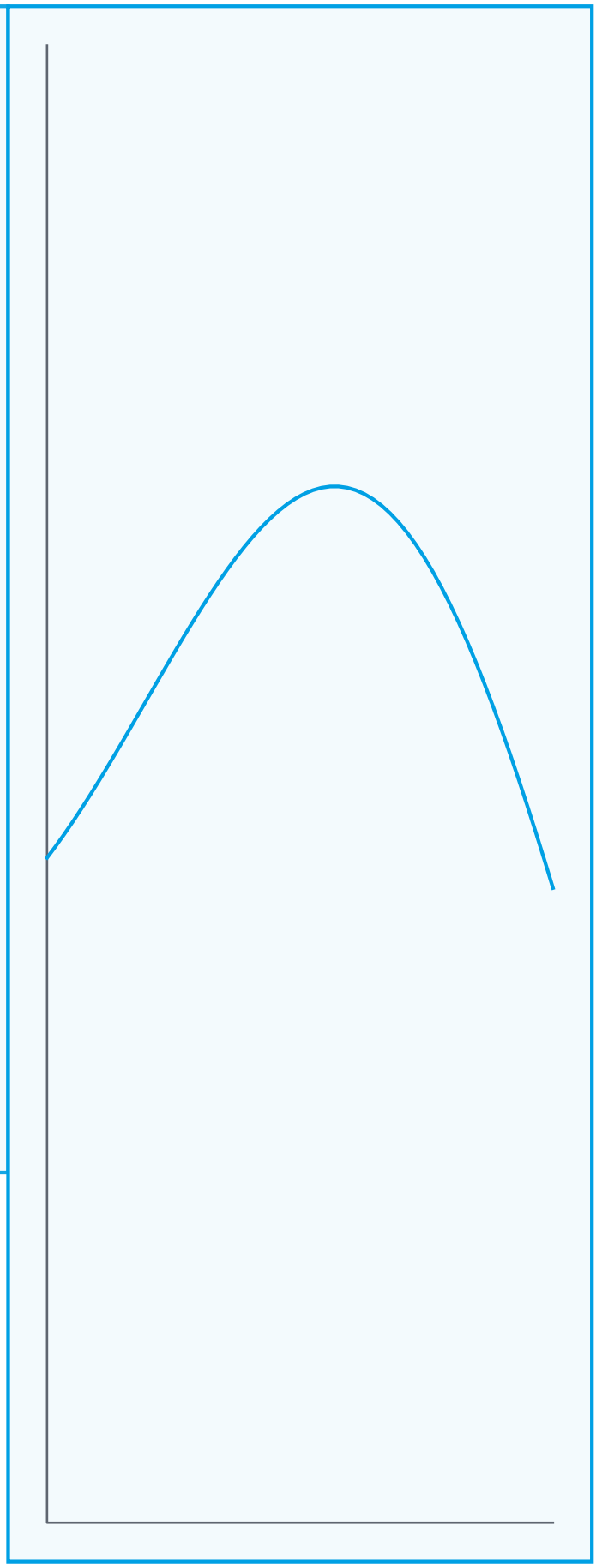


Figure 11 - single_long (9 x 24 cm)

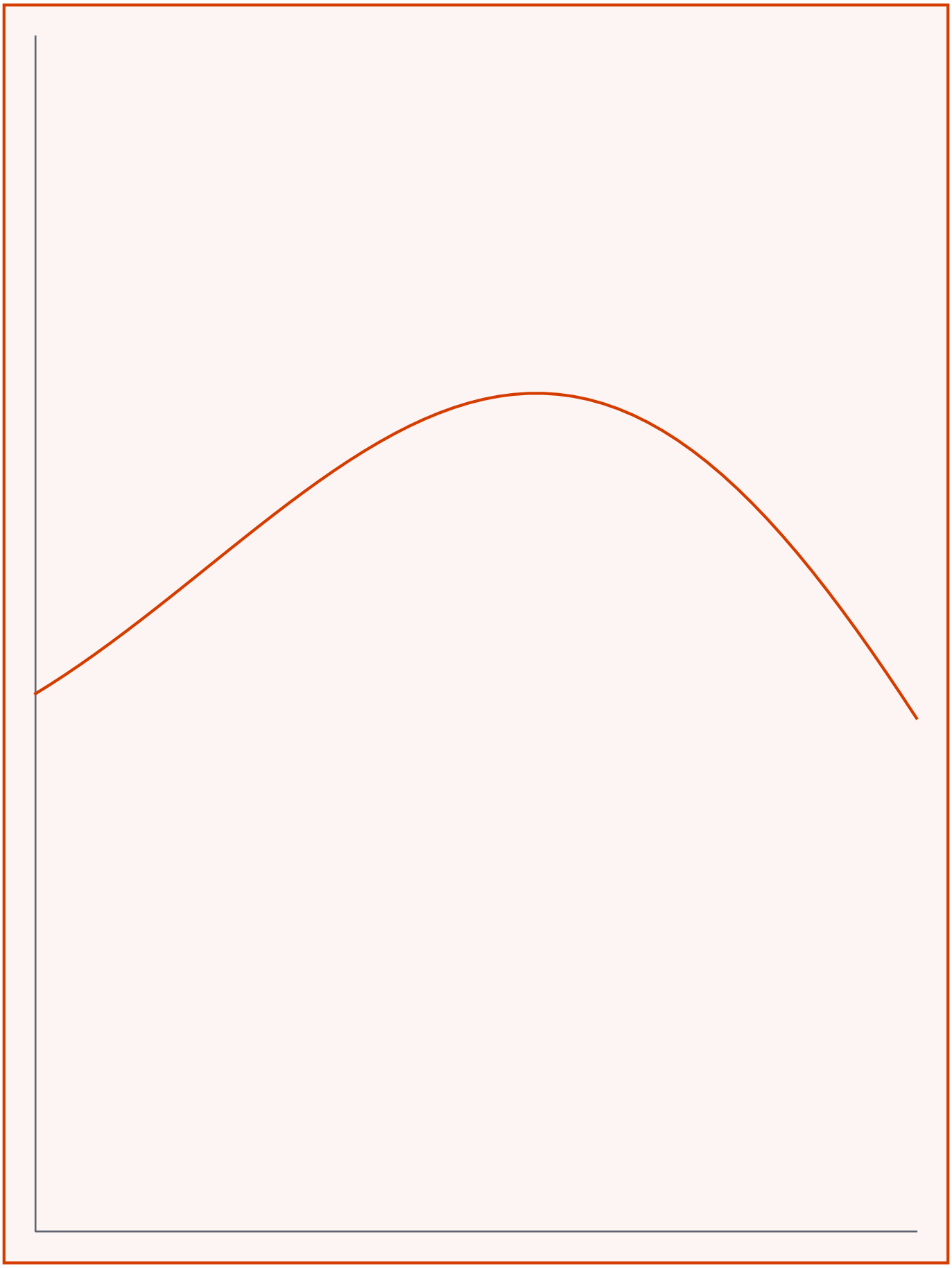
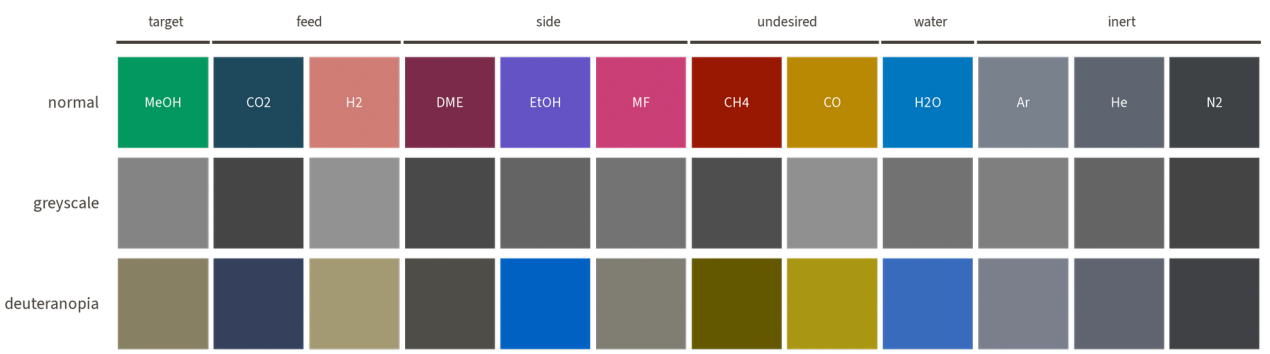


Figure 12 - double_long (18 x 24 cm)

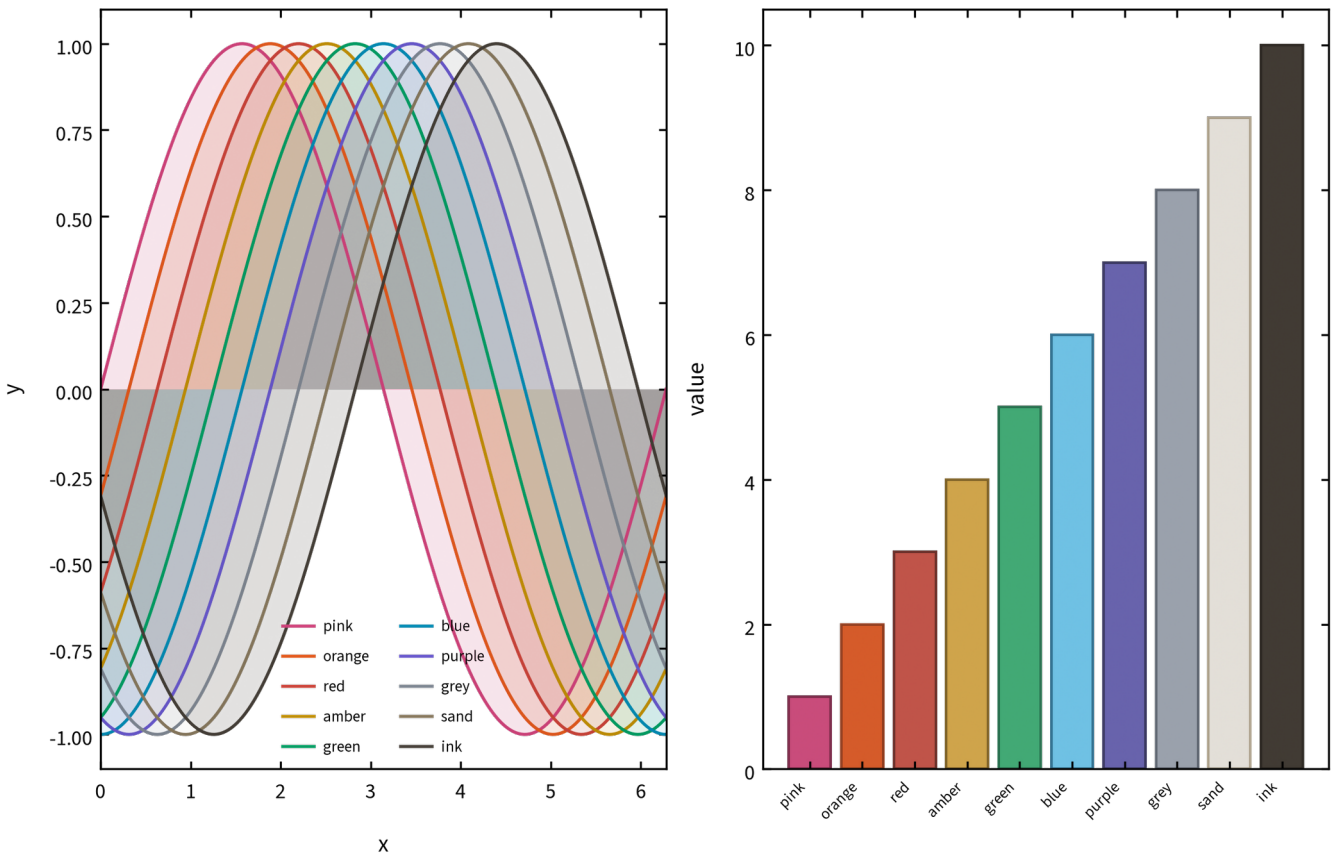
Ten hues, four fill tones each for drawings and one line tone for plots. Species reuse the hues: green target, blue feed, purple and pink side products, red and amber undesired, grey inert.





The species palette under normal vision, in greyscale and under deuteranopia.

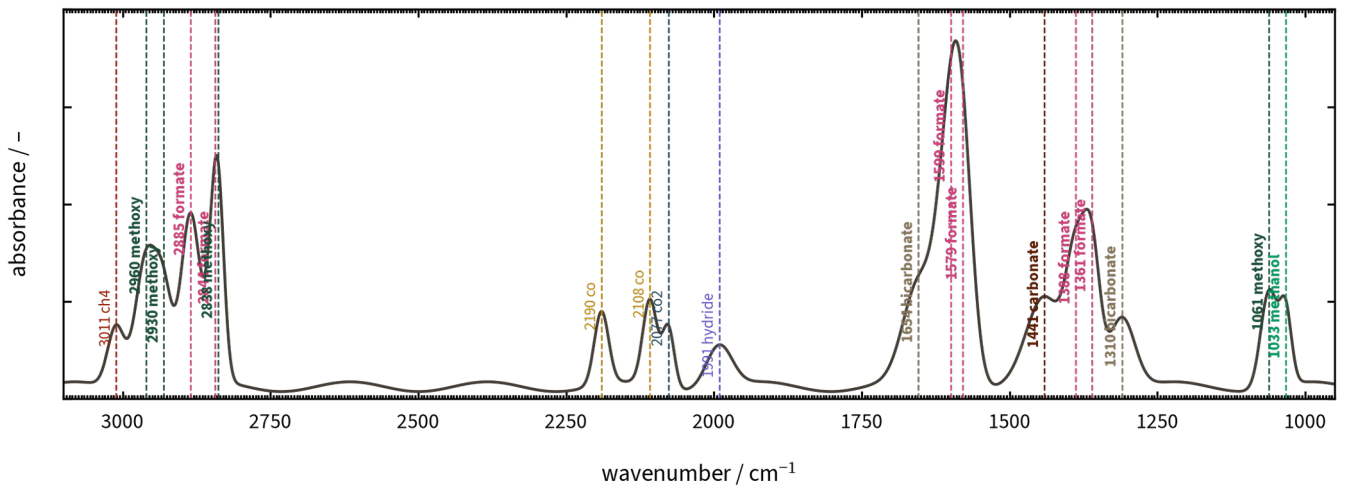
Lines take the line tone, bands and patches the base tone: a thin stroke reads less saturated than the same colour as an area, so the two tones differ in chroma, not in hue.



Nature's own palette, for a conceptual drawing that should look like a Nature figure. Background families for context, accents for the action, the extended set to group.

	0	1	2	3	4	5
background **stone**	#f7f5ef	#e4e0ce	#c6c2a6	#a8a386	#888366	#625f4a
grey	#e6e6ed	#c8ceda	#99a3b4	#6f7b91	#49566d	#253247
main accents **red**	#fad0ce	#eda4a7	#de6866	#c93e3f	#9c2826	#741915
blue	#c8e7fb	#9dcbec	#5799d1	#0272b2	#014e91	#012c5c
yellow	#fff1c3	#f6dc8a	#ebc850	#cca02c	#9c7717	#6b5513
extended **olive**	#f4f1b3	#dfdc67	#c9c700	#9ba415	#66771e	#36461a
green	#d9e8c6	#a0ca79	#62b347	#459434	#227130	#163b1c
teal	#cce7ee	#97d2d4	#4bbcbd	#019aa3	#016879	#00394e
blue	#c8e7fb	#9dcbec	#5799d1	#0272b2	#014e91	#012c5c
purple	#ebd6e9	#d3aad1	#bb7cb4	#a84e94	#792c74	#481951
red	#fad0ce	#eda4a7	#de6866	#c93e3f	#9c2826	#741915
orange	#fde0bd	#fbc07f	#f59a45	#ec6f00	#b74f06	#843200
yellow	#fff1c3	#f6dc8a	#ebc850	#cca02c	#9c7717	#6b5513
skin	#f8e6d7	#ddbda3	#bf997d	#936a57	#755040	#442d1f

A band is coloured by the species it belongs to, with the species' own colour, so a molecule looks the same in the atlas, in a spectrum and in an MS trace. Progress along the pathway is carried by lightness within a family, not by a hue sequence.



band kinds

- formate
- carbonate
- bicarbonate
- methoxy
- methanol
- co2
- co
- ch4
- water
- hydride
- unassigned

feed halves in a switching experiment

- h2
- co2
- co2 + h2
- inert

Figure text is Source Sans 3 and figure maths Source Serif 4, both bundled, so a figure looks the same on every machine. Below, the same line in the fonts a reader would get by default, all at 9 pt and at true size.

text

Source Sans 3	*phd_ui, print and slides*	temperature / K absorbance / – 1000/T 0.25 0.50 250 °C
Segoe UI	*the slide body font so far*	temperature / K absorbance / – 1000/T 0.25 0.50 250 °C
Calibri	*PowerPoint's default*	temperature / K absorbance / – 1000/T 0.25 0.50 250 °C
DejaVu Sans	*matplotlib's default*	temperature / K absorbance / – 1000/T 0.25 0.50 250 °C

maths

custom	*Source Serif 4, phd_ui*	$CO_2 + 3\,H_2$ S_{MeOH} $k = A\exp(-E_a/RT)$ $\frac{1000}{T}$ $\nu_{as}(OCO)$
dejavusans	*matplotlib's default*	$CO_2 + 3\,H_2$ S_{MeOH} $k = A\exp(-E_a/RT)$ $\frac{1000}{T}$ $\nu_{as}(OCO)$
stix	*Times-like, as in most journals*	$CO_2 + 3\,H_2$ S_{MeOH} $k = A\exp(-E_a/RT)$ $\frac{1000}{T}$ $\nu_{as}(OCO)$
cm	*Computer Modern, LaTeX*	$CO_2 + 3\,H_2$ S_{MeOH} $k = A\exp(-E_a/RT)$ $\frac{1000}{T}$ $\nu_{as}(OCO)$

sizes at true scale

body text, 10 pt serif	*the thesis body, what a figure is read beside*
axis label, 9 pt	*axes.labelsize*
tick label, legend, 8 pt	*font.size, legend.fontsize, the publisher window is 5 to 9 pt*
caption, 8 pt serif	*set by the document, not by the figure*

on a slide every figure point size is doubled: 16 pt ticks, 18 pt labels, beside 18 to 24 pt bullets

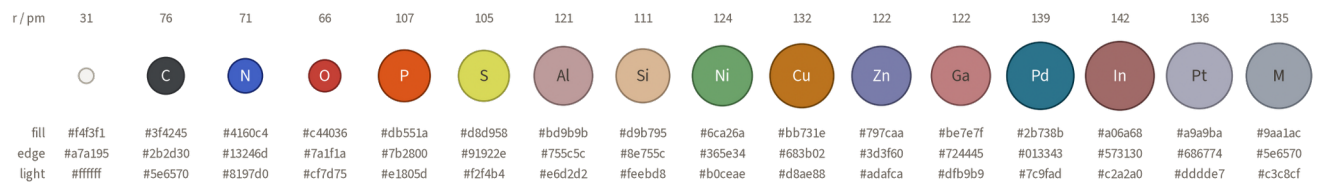
guideline

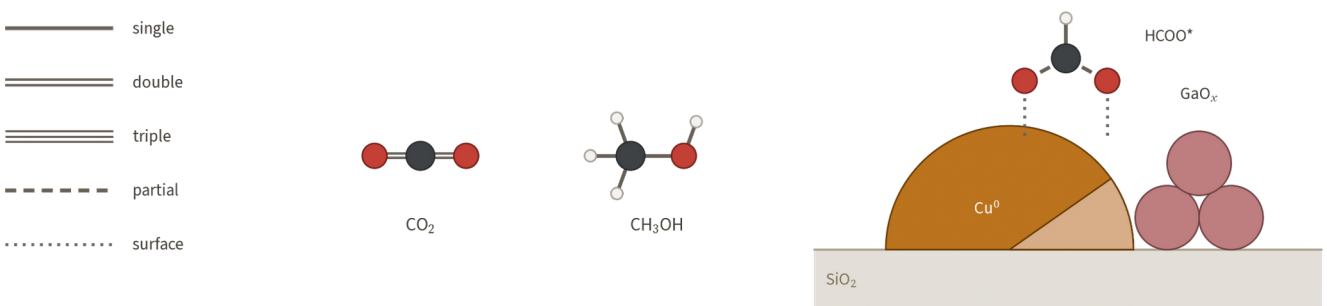
One sans for all figure and slide text: Source Sans 3. Hierarchy by size and weight, never a second sans.
Maths matches the body text beside it: Source Serif 4 in print figures and the thesis, Source Sans 3 italic on slides.
Two families per document at most, from one design family. Words and units upright, quantities italic.
Mathtext, not usetex. Slides move from Segoe UI to Source Sans 3, embedded in the pptx.
Thesis: LuaLaTeX, Source Serif 4 body, Source Sans 3 headings and captions, unicode-math with
Source Serif 4 letters over STIX Two Math symbols. Preamble and to-dos in phd_ui CLAUDE.md, Typography.

Wong (2011): up to 8 % of men are red-green colour blind. Vary lightness and saturation, not only hue; treat any colour with red and green in it as suspect; label rather than rely on colour alone. His palette, for when a figure has to follow it:



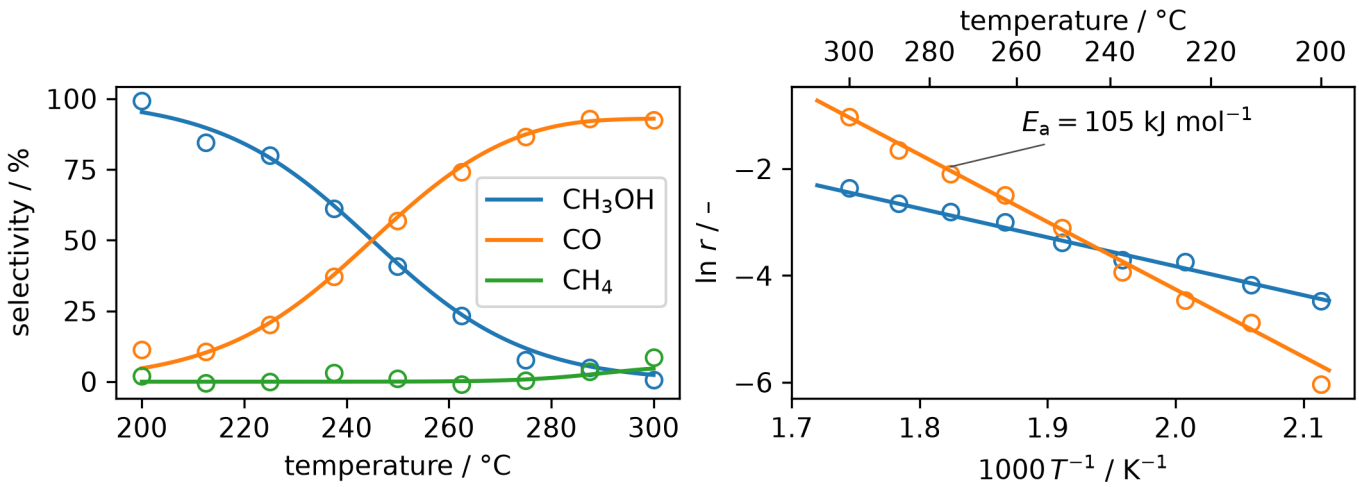
Atoms: CPK hues toned to the scheme, flat, carbon dark grey. Radius = covalent radius; ball and stick at half of it with true bond lengths. Symbols inside only from 4 mm.



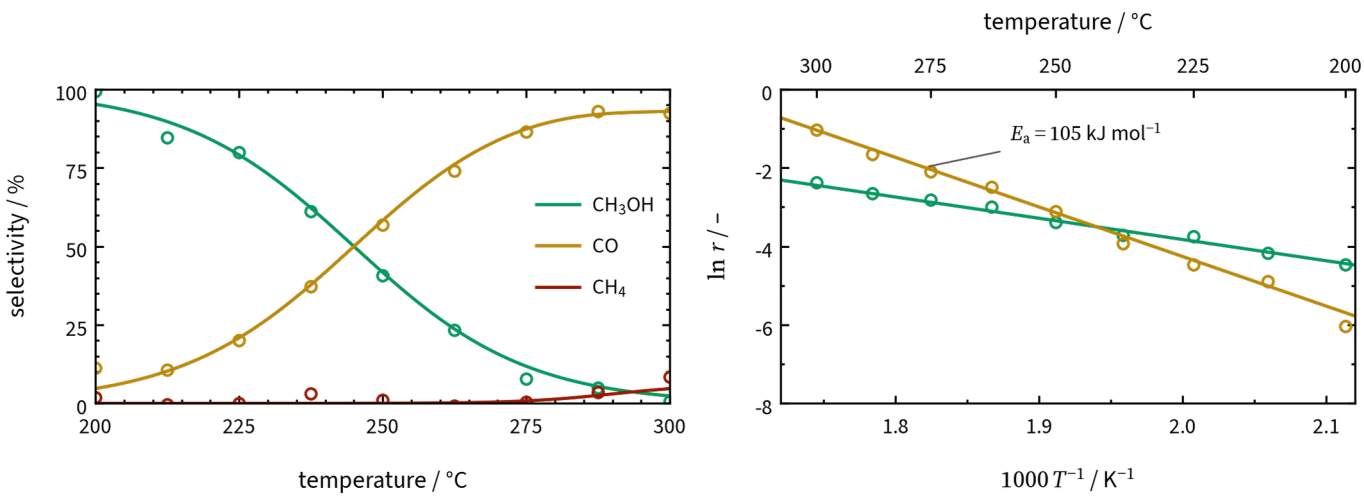


One figure, twice: same data, same label strings, same 18 x 6.5 cm canvas (double_single). Only the style differs.
Both are placed at true size, so the point sizes on this page are the ones a reader of the paper would see.

matplotlib defaults



phd_ui, double_single



what changes

Size: 10 pt everywhere is too large at 18 cm. phd_ui sets 8 pt ticks and legend, 9 pt axis labels, 2x on slide keys.

Font: DejaVu Sans for text and maths becomes Source Sans 3 for text, Source Serif 4 for maths.

Colour: tab10 becomes the scheme line tones; species take SPECIES[name].color, the same in every figure.

Ticks: inward on all four spines; limits and spacing chosen together with set_locators, a tick on each end.

Limits: selectivity runs 0 to 100 %, not over the data range. Legend without a frame.

Second axis: add_inverse_temperature_axis replaces a hand-written secondary_xaxis with lambdas.

matplotlib has no mhchem, so \ce{} does not exist in a figure. Each row shows a way of writing a label, the source string, and what it renders to under the print preset.

plain text	`"CO2"`	CO2
unicode subscript	`"CO₂"`	CO_2
maths roman (print: serif)	`r"$\mathrm{CO_2}$"`	$\mathrm{CO_2}$
maths sans (follows the text)	`r"$\mathsf{CO_2}$"`	$\mathsf{CO_2}$
ion, oxidation state	`r"$\mathsf{Al^{3+}}$, $\mathsf{Cu^0}$"`	$\mathsf{Al^{3+}}$, $\mathsf{Cu^0}$
reaction	`r"$\mathsf{CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O}$"`	$\mathsf{CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O}$
quantity, upright subscript	`r"$S_\mathrm{MeOH}$ / %"`	S_MeOH / %
unit exponent, maths	`r"kJ mol$^{-1}$"`	kJ mol^{-1}
unit exponent, sans	`r"$\mathsf{kJ\,mol^{-1}}$"`	$\mathsf{kJ\,mol^{-1}}$
unit exponent, unicode	`"kJ mol⁻¹"`	kJ mol⁻¹

guideline

A species standing as a name (legend, annotation) is maths sans: $\mathsf{CO_2}$, $\mathsf{Al^{3+}}$.
It resolves to Source Sans 3, so the formula matches the words beside it. Reactions the same way.
A species as the subscript of a quantity stays \mathrm: S_MeOH, E_a, matching the italic symbol.
Not $\mathrm{CO_2}$ for a bare species: on print presets it is set in Source Serif 4, serif inside sans text.
Quantities go in maths so they are italic: T, $1000\,T^{-1}$ / K^{-1}, $\ln\, r$ / –.
Units are text with a maths exponent, kJ mol^{-1}, or all sans, $\mathsf{kJ\,mol^{-1}}$. Mathtext, never usetex.